Angular Under the Hood

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SAP Commissions Tech Meetup – 11/04/2020

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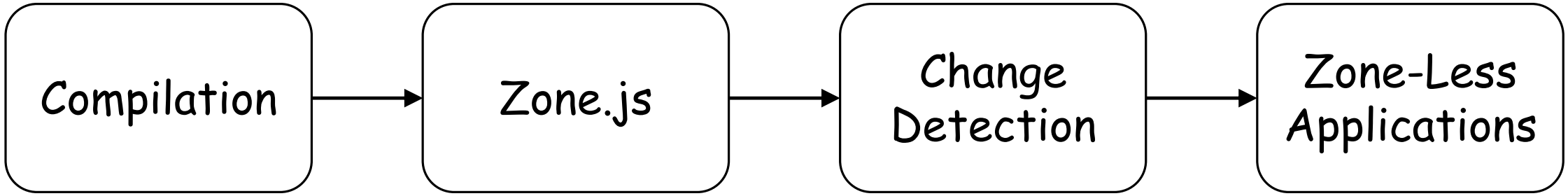
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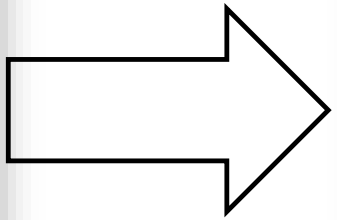
Compilation

```
import { Component } from '@angular/core';

@Component({
  selector: 'app-home',
  template: `
    <h1>Home Component</h1>
    <h3>{{ subtitle }}</h3>
  `
})
export class HomeComponent {
  subtitle = 'Hello World';
}
```



```
class HomeComponent {
  constructor() {
    this.subtitle = 'Hello World';
  }
}
HomeComponent.ɵfac = function HomeComponent_Factory(t) {
  return new (t || HomeComponent)();
};
HomeComponent.ɵcmp = ɵɵdefineComponent({
  type: HomeComponent,
  selectors: [["app-home"]],
  decls: 4,
  vars: 1,
  template: function HomeComponent_Template(rf, ctx) {
    if (rf & 1) { // ɵRenderFlags.Create
      ɵɵelementStart(0, "h1"); // <h1>
      ɵɵtext(1, "Home Component"); // Home Component
      ɵɵelementEnd(); // </h1>
      ɵɵelementStart(2, "h3"); // <h3>
      ɵɵtext(3); // {{ subtitle }}
      ɵɵelementEnd(); // </h3>
    }
    if (rf & 2) { // ɵRenderFlags.Update
      ɵɵadvance(3); // advance three times to '{{ subtitle }}' node
      ɵɵtextInterpolate(ctx.subtitle); // update '{{ subtitle }}' value
    }
  },
  encapsulation: 2
});
```

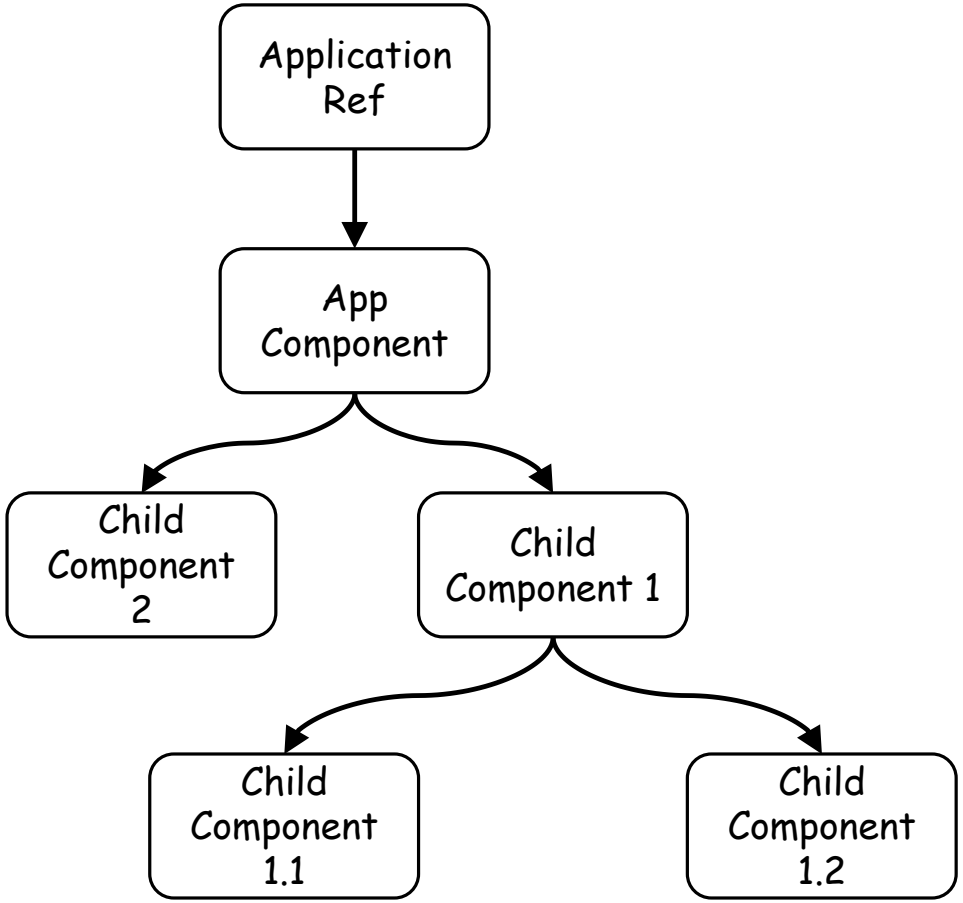
What is Zone.js?

- Executed before the Angular application is bootstrapped
- Tells Angular when to run the change detection mechanism
- Monkey patches asynchronous browser APIs

```
Angular is running in development mode. Call enableProdMode() to enable production mode.
[WDS] Live Reloading enabled.
> Object.keys(setTimeout);
<· ▸["__zone_symbol__OriginalDelegate"]
> Object.keys(setInterval);
<· ▸["__zone_symbol__OriginalDelegate"]
> Object.keys(EventTarget.prototype.addEventListener);
<· ▸["__zone_symbol__OriginalDelegate"]
> Object.keys(Promise);
<· ▸["__zone_symbol__uncaughtPromiseErrors"]
```

Change Detection

```
@Injectable()
export class ApplicationRef {
  constructor(private _zone: NgZone) {
    this._zone.onMicrotaskEmpty.subscribe(() =>
      this._zone.run(() => this.tick());
    );
  }
}
```



Zone.js Downsides

- Not tree shakable (uncompressed size >100kB)
- Slower application bootstrap speed
- Unnecessary change detection triggering
- Magic
- Cannot monkey patch native async/await (ES2017)

Zone-Less Applications

- Trigger change detection only when necessary

```
// pollyfills.ts
// import 'zone.js/dist/zone';

// main.ts
platformBrowserDynamic()
  .bootstrapModule(AppModule, { ngZone: 'noop' }) //
  .catch(err => console.error(err));
```

```
import { Component, ɵmarkDirty } from '@angular/core';

@Component({
  selector: 'app-counter',
  template: `
    <h1>Counter</h1>
    <div>
      <button (click)="onIncrement()">+</button>
      <span>{{ count }}</span>
      <button (click)="onDecrement()">-</button>
    </div>
  `,
})
export class CounterComponent {
  count = 0;

  onIncrement(): void {
    this.count++;
    ɵmarkDirty(this);
  }

  onDecrement(): void {
    this.count--;
    ɵmarkDirty(this);
  }
}
```

How to trigger change detection automatically?

```angular-core.component.ts
import { Component } from '@angular/core';
import { BehaviorSubject } from 'rxjs';

@Component({
  selector: 'app-counter',
  template: `
    <h1>Counter</h1>
    <div>
      <button (click)="onIncrement()">+</button>
      <span>{{ count | zoneLessAsync }}</span> <!-- 👈 -->
      <button (click)="onDecrement()">-</button>
    </div>
  `,
})
export class CounterComponent {
  count = new BehaviorSubject(0);

  onIncrement(): void {
    this.count.next(this.count.value + 1);
  }

  onDecrement(): void {
    this.count.next(this.count.value - 1);
  }
}
```

Demo

Thanks!

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